

Code Review Report

Role-Based Chatbot Task Assignment

Project: Linkific – HR Automation Platform

Feature Reviewed: Role-Based Chatbot Task Assignment

1. Feature Overview

The Role-Based Chatbot Task Assignment feature allows an authorized user, such as an Admin, to assign tasks to employees using the chatbot. The chatbot processes the user's request, identifies the required task and employee, checks the user's role and permissions, and performs the task assignment through the backend. This reduces manual task assignment and allows administrators to interact with the HR system using natural language.

2. Code Review Objective

The implemented feature was reviewed to identify areas for improvement in code readability, modularity, naming conventions, documentation, error handling, maintainability, and security.

3. Code Review Findings

3.1 Code Readability

The feature logic can become difficult to understand when chatbot processing, permission checking, employee identification, and task creation are handled together. The logic should be organized into clear steps and functions so that another developer can quickly understand the implementation.

3.2 Modularity

Functionality such as permission validation, employee lookup, and task creation should be separated into reusable service functions. Suggested functions include `validate_task_assignment_permission()`, `find_employee()`, `create_task()`, and `assign_task_to_employee()`.

3.3 Naming Conventions

Use descriptive variable and function names such as `employee_id`, `task_description`, `assigned_by`, `assignment_result`, `assign_task_to_employee()`, `validate_admin_permission()`, and `get_employee_by_id()`. Consistent naming improves readability and reduces confusion.

3.4 Documentation

Important chatbot tools and backend functions should clearly explain their purpose, parameters, return values, and possible errors. Docstrings should be added to important functions and API/tool documentation should describe expected inputs and outputs.

3.5 Error Handling

The feature should handle invalid requests and unexpected situations gracefully, including a missing employee, empty task description, insufficient permission, invalid employee ID, database failure, or task creation failure. Users should receive meaningful error messages instead of internal exceptions.

3.6 Authorization and Security

Because this is a role-based feature, authorization must be enforced by the backend and should not depend only on the chatbot's interpretation. The system should independently verify authentication, role, and required permission before creating a task.

4. Improvements Made / Recommended

Area	Improvement
Readability	Simplified and organized feature logic.
Modularity	Separated permission, employee lookup, and task assignment logic.
Naming	Used descriptive function and variable names.
Documentation	Added docstrings and clearer explanations.
Error Handling	Added validation and meaningful error responses.
Security	Ensured authorization before task assignment.
Maintainability	Reduced tightly coupled logic.

5. Expected Improved Flow

Admin → Chatbot → Intent/Request Processing → Permission Validation → Employee Validation → Task Creation → Confirmation

Example

Admin: "Assign a task to Employee 12 to complete the onboarding documentation."

- Identify the requested action as task assignment.
- Identify Employee 12.
- Extract the task description.
- Verify that the admin has permission to assign tasks.
- Verify that Employee 12 exists.
- Create the task.
- Return a confirmation to the admin.

6. Conclusion

The Role-Based Chatbot Task Assignment feature provides a convenient way for authorized users to assign employee tasks through natural language. The code review identified opportunities to improve readability, modularity, naming conventions, documentation, error handling, maintainability, and authorization. Applying these improvements makes the feature easier to maintain, safer to use, and more suitable for production deployment.